

Keele energy challenge

SEND campus energy data | 17 September 2026

github.com/SENDatKeele/ERA-event

Start with README.md and TASK.md. CSV files and the data dictionary are in data/.

When does generation exceed demand?

The dataset

Three annual CSV files cover 2023, 2024 and 2025. Each row covers five minutes.

Column	Plain meaning	Unit
DateTime	Date and time	Date/time
power-con-ave	Campus electricity consumption	kW
power-gen-wt-ave	Wind-turbine generation	kW
power-gen-pv-ave	Solar generation	kW

2023 is reviewed and repaired. 2024 and 2025 are reconstructed. Compare years with care.

Power and energy

Each row is a five-minute average power reading. Energy accumulates over time.

Measure	Plain meaning	Unit
Power	Rate of electricity use or generation	kW
Energy	Power accumulated over time	MWh
Balance	Generation minus consumption	kW
Excess	The positive part of the balance	kW

For five-minute data: energy in MWh = sum of kW readings / 12,000.

Data exploration

01	Open one annual CSV. Check the columns, units and dates.
02	Plot consumption, wind and solar for one day or week.
03	Note one pattern that looks reasonable.
04	Note one feature you would check before relying on it.

Use VS Plotter by Rafael Arvelo. No coding needed. Steps are in TASK.md.

Challenge options

02	Compare years: how do demand and renewable generation change?
03	Find excess: when, how often and for how long?
04	Audit zeros: which look expected, and which need checking?
05	Advise Keele: which flexible response fits a recurring surplus?

State your assumptions. A careful limitation is part of a strong answer.

Group presentation

01	One clearly labelled plot
02	One key finding or question from the data
03	Why it matters or how it could be used
04	Any assumption or limitation needed to understand the finding

One slide, about two minutes. Focus on one finding.

Evidence and limits

Show the finding, its limits and the next check.

Sharing or continuing your work? Credit Keele's SEND team and cite the data repository and paper. Describe your own contribution.

Full citation and example wording: ACKNOWLEDGEMENTS.md